

# SpaceX Falcon 9 First Stage Landing Prediction

IBM Data Science Professional Certificate

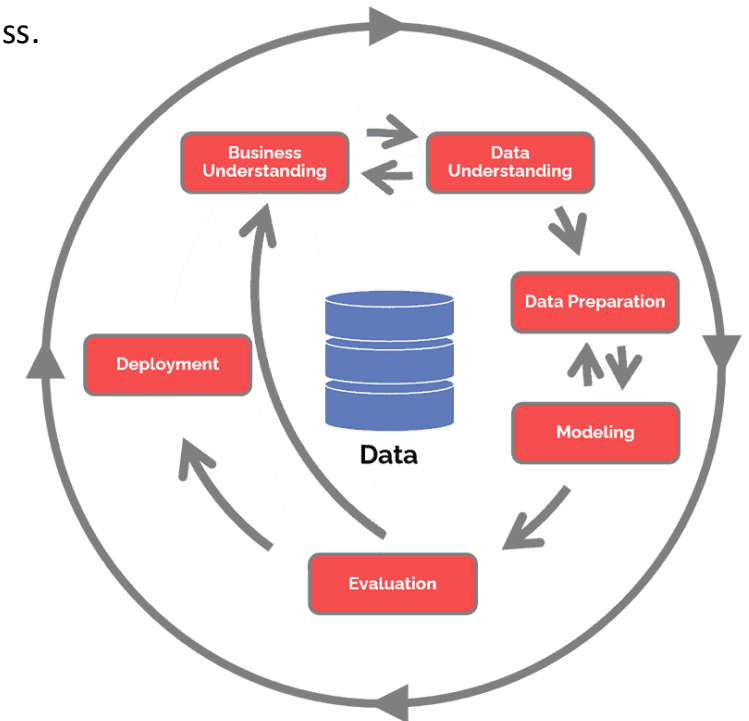
Applied Data Science Capstone

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# Executive Summary

- This project focuses on predicting the successful landing of the SpaceX Falcon 9 first stage using data science techniques.
- The project workflow includes data collection, data cleaning, exploratory data analysis, interactive visualization, and machine learning classification.
- Launch data was collected from the SpaceX API and public sources. Python, SQL, Folium, and Plotly Dash were used for analysis and visualization.
- Multiple machine learning models were developed to predict Falcon 9 first-stage landing success.



# Introduction

- SpaceX Falcon 9 is a reusable rocket system designed to reduce launch costs.
- Successful first-stage landing allows booster reuse and improves mission efficiency.
- Historical launch data can be analyzed to identify factors affecting landing success.

## Project Objectives:

- Collect launch data
- Clean and prepare data
- Perform EDA
- Create visualizations
- Build prediction models



# Data Collection Methodology

## Data Sources:

- SpaceX REST API
- Public web sources

## Tools Used:

- Python Requests
- Pandas
- Web Scraping

## Collected Features:

- Launch site
- Payload mass
- Orbit type
- Booster version
- Landing outcome

# Data Collection Evidence

The collected launch data was converted into a structured dataset using Python and Pandas.

The dataset contains important mission characteristics used for further analysis.

## Task 1: Request and parse the SpaceX launch data using the GET request

To make the requested JSON results more consistent, we will use the following static response object for this project:

```
static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.js'
```

We should see that the request was successful with the 200 status response code

```
response=requests.get(static_json_url)
```

```
response.status_code
```

```
200
```

Now we decode the response content as a Json using `.json()` and turn it into a Pandas dataframe using `.json_normalize()`

```
data = pd.json_normalize(response.json())
```

Using the dataframe `data` print the first 5 rows

```
# Get the head of the dataframe  
data.head()
```

|   | static_fire_date_utc     | static_fire_date_unix | tbd   | net   | window | rocket                   | success | details                                                                                                                                                                              | crew | ships | capsules          |
|---|--------------------------|-----------------------|-------|-------|--------|--------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|-------------------|
| 0 | 2006-03-17T00:00:00.000Z | 1.142554e+09          | False | False | 0.0    | 5e9d0d95eda69955f709d1eb | False   | Engine failure at 33 seconds and loss of vehicle                                                                                                                                     | []   | []    | [5eb0e4b5b6c3bb0C |
| 1 | NaN                      | NaN                   | False | False | 0.0    | 5e9d0d95eda69955f709d1eb | False   | Successful first stage burn and transition to second stage, maximum altitude 289 km, Premature engine shutdown at T+7 min 30 s, Failed to reach orbit, Failed to recover first stage | []   | []    | [5eb0e4b5b6c3bb0C |

# Data Wrangling Methodology

Data preprocessing steps:

- Identified missing values
- Removed unnecessary columns
- Converted data types
- Processed categorical variables
- Prepared dataset for machine learning

# Data Cleaning Results

The cleaned dataset improved data quality and ensured reliable analysis and model development.

VIEW LOGS | CODE | MINIMAP

Identify and calculate the percentage of the missing values in each attribute

```
(df.isnull().sum()/len(df))*100
```

[3] ✓ 0.0s Python

```
... FlightNumber    0.000000
Date              0.000000
BoosterVersion    0.000000
PayloadMass       0.000000
Orbit             0.000000
LaunchSite        0.000000
Outcome           0.000000
Flights           0.000000
GridFins          0.000000
Reused            0.000000
Legs              0.000000
LandingPad        28.888889
Block             0.000000
ReusedCount       0.000000
Serial            0.000000
Longitude         0.000000
Latitude          0.000000
dtype: float64
```

handling missing values

handling missing values

```
df["LandingPad"] = df["LandingPad"].fillna("Unknown")
(df.isnull().sum()/len(df))*100
```

[4] ✓ 0.0s Python

```
... FlightNumber    0.0
Date              0.0
BoosterVersion    0.0
PayloadMass       0.0
Orbit             0.0
LaunchSite        0.0
Outcome           0.0
Flights           0.0
GridFins          0.0
Reused            0.0
Legs              0.0
LandingPad        0.0
Block             0.0
ReusedCount       0.0
Serial            0.0
Longitude         0.0
Latitude          0.0
dtype: float64
```

# EDA and Data Visualization Methodology

EDA was performed to identify relationships between launch characteristics and landing success.

## Tools Used:

- Pandas
- Matplotlib
- Seaborn
- SQL

## Analysis:

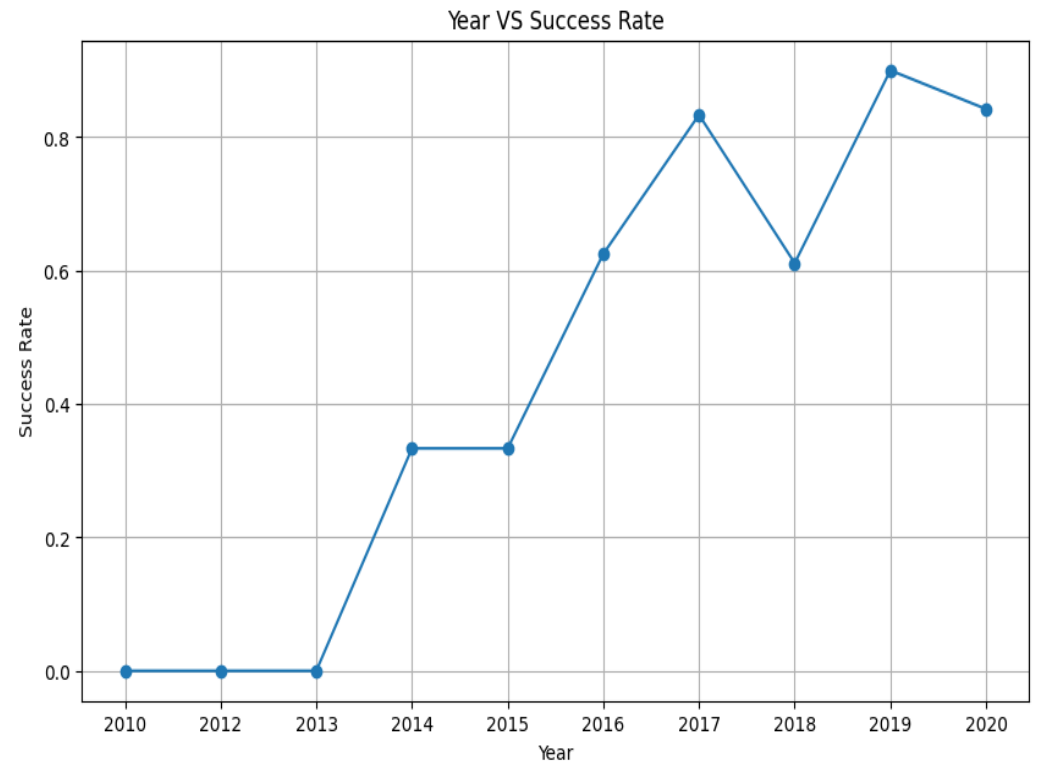
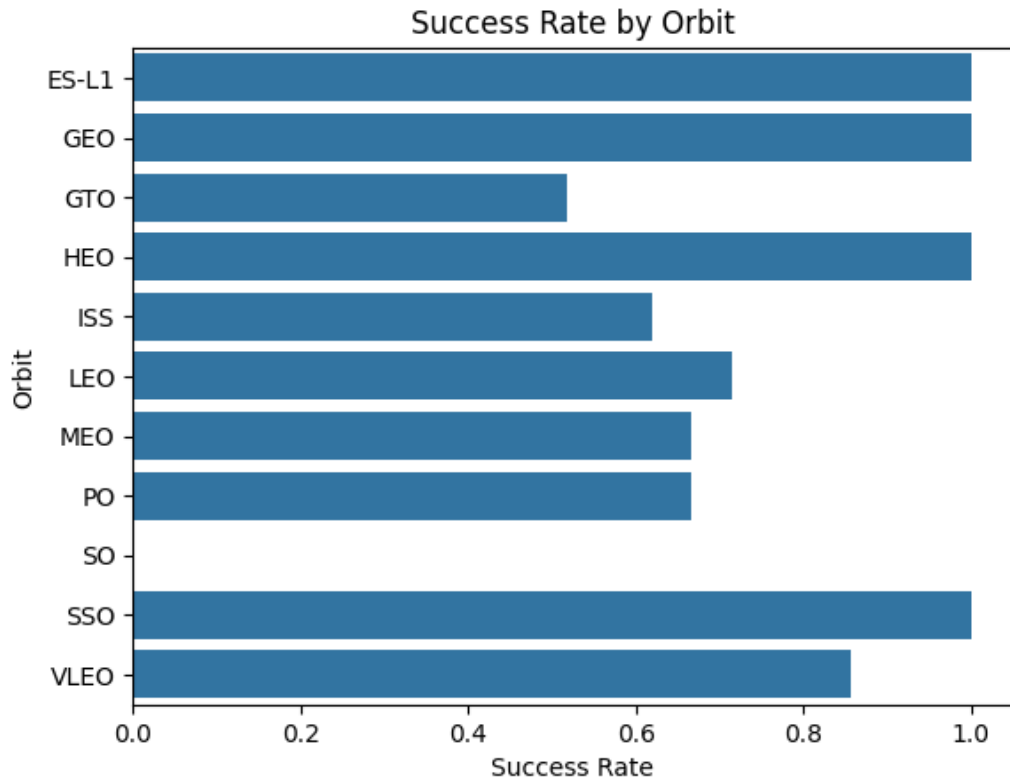
- Launch sites
- Payload mass
- Flight number
- Orbit type



# Exploratory Data Analysis Results

Key observations:

- Launch sites showed different success rates.
- Payload characteristics influenced landing outcomes.
- Success patterns changed with flight experience.



# EDA Using SQL Queries

SQL queries were used to analyze launch data.

Analysis included:

- Launch site performance
- Number of successful launches
- Payload information
- Mission characteristics

# SQL Query Results

SQL analysis helped identify patterns in launch frequency and mission success.

## Query

Click to add a breakpoint

```
SELECT Mission_Outcome, COUNT(*) AS Total
FROM SPACEXTBL
GROUP BY Mission_Outcome;
""",conn)
```

## Result

|   | Mission_Outcome                  | Total |
|---|----------------------------------|-------|
| 0 | Failure (in flight)              | 1     |
| 1 | Success                          | 98    |
| 2 | Success                          | 1     |
| 3 | Success (payload status unclear) | 1     |

## Query

Click to add a breakpoint

```
SELECT
    Landing_Outcome,
    COUNT(*) AS Count
FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Count DESC;
""", conn)
```

## Result

|   | Landing_Outcome        | Count |
|---|------------------------|-------|
| 0 | No attempt             | 10    |
| 1 | Success (drone ship)   | 5     |
| 2 | Failure (drone ship)   | 5     |
| 3 | Success (ground pad)   | 3     |
| 4 | Controlled (ocean)     | 3     |
| 5 | Uncontrolled (ocean)   | 2     |
| 6 | Failure (parachute)    | 2     |
| 7 | Precluded (drone ship) | 1     |

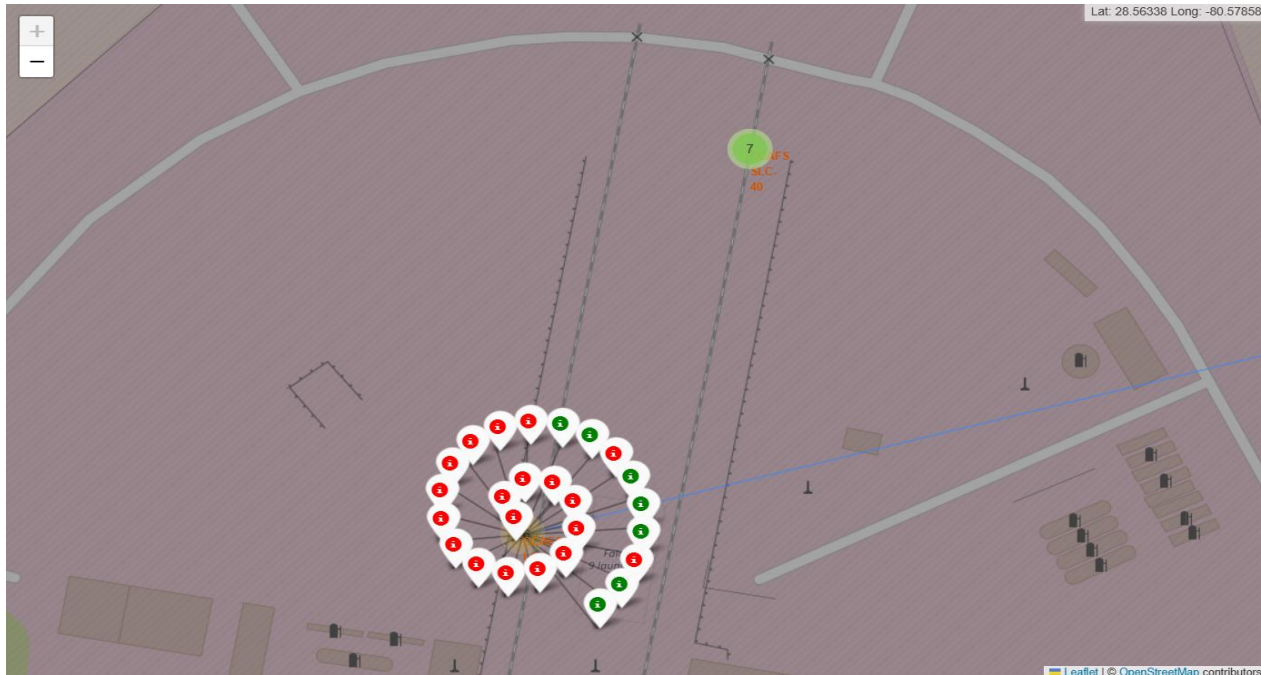
# Launch Site Analysis Using Folium

Folium was used to create an interactive map of SpaceX launch locations.

The map helps analyze:

- Geographic distribution
- Launch site locations
- Success patterns

Polyline between launched site and coastline point



# Interactive Dashboard Using Plotly Dash

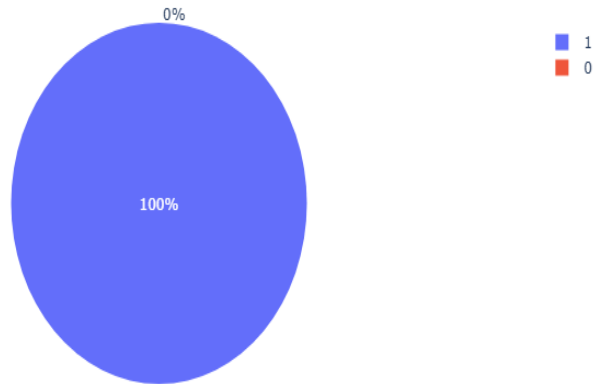
A Plotly Dash dashboard was developed for interactive exploration.

Features:

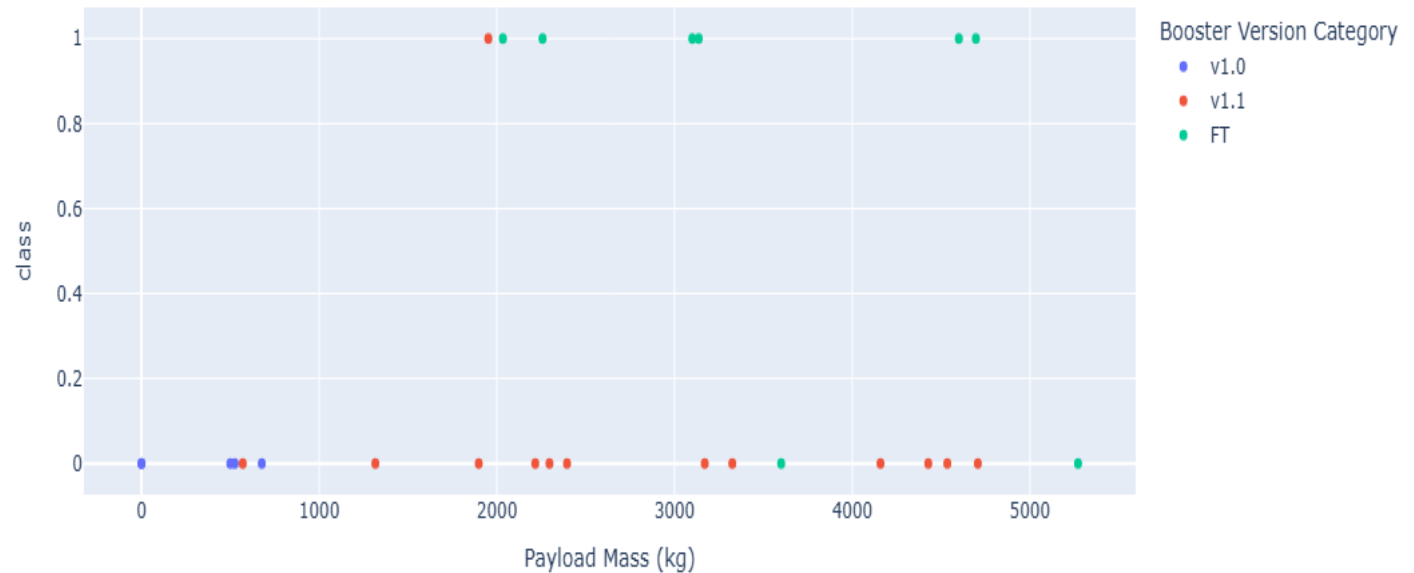
- Launch site filtering
- Payload selection
- Dynamic success visualization

Pie charts and scatter plot of CCAFS LC-40

Success Rate for CCAFS LC-40



Payload Mass vs Launch Outcome



# Predictive Analysis (Classification)

Machine learning models were developed to predict landing success.

Models:

- Logistic Regression
- Support Vector Machine
- Decision Tree
- K-Nearest Neighbors

Steps:

- Train-test split
- Feature scaling
- Model evaluation

# Predictive Analysis Results

Model performance was compared using accuracy scores.

The best-performing model was selected based on prediction accuracy.

| Model                        | Accuracy |
|------------------------------|----------|
| Logistic Regression          | 0.83     |
| Support Vector Machine (SVM) | 0.88     |
| Decision Tree                | 0.78     |
| K-Nearest Neighbors (KNN)    | 0.83     |

# Conclusion

- Completed the complete data science workflow.
- Collected and cleaned SpaceX launch data.
- Performed EDA using Python and SQL.
- Created interactive visualizations.
- Developed machine learning models for landing prediction.



# Git hub

- <https://github.com/kulbhushanranaware/IBM-Data-Science-Capstone>

# Thank You

## Questions & Discussion